

01. Business Requirements Document (BRD)

Project	estateCRM
Version	0.6
Date	2026-09-24
Author	Sarang Bondre (product owner), drafted with Claude; v0.4 to v0.6 revised with Claude from the 11 Estates operating system sessions
Status	AWAITING APPROVAL

Change log

Version	Change
0.1	Initial draft
0.2	Answered OQ-1 (internal first, SaaS later), OQ-2 (India, Mumbai first), OQ-3 (volume), OQ-4 (dedup is a hard problem to design), OQ-5 (Lead deferred), OQ-8 (Listings API scope). Scope, metrics, constraints and risks updated to match.
0.3	Company name recorded as "11estate" (OQ-18). Corrected in 0.4.
0.4	Company name corrected to 11 Estates . Document restructured around the demand and supply journeys. New sections: data model (4), status model, life curve and publication rules (4), supply journey (5), demand journey (6), matching rules (7) and dashboards (8). Brought into scope: deal stages up to Closed, Enquiry, single record entry, publication levels, project microsites and anonymous demand posts. OQ-12, OQ-14, OQ-15, OQ-16 and OQ-17 answered. File upload remains the only ingestion channel. Stack and success metrics unchanged from 0.3.
0.5	A-2 confirmed: Phase 1 needs both file upload and manual entry and updates in the app. A-17 confirmed: lead and enquiry exports are uploaded daily. Connectors confirmed for the next phase; OQ-7 reworded to match.
0.6	Deal vocabulary standardized (2026-09-24). New 4.2 Record classification and deal vocabulary: record_scope, deal_type (Sale, Lease, JV, Pagdi), market, four segments with property types, side with side_evidence, deal tags, party fields and generated display labels. Data model, intake, journeys, matching, dashboards, scope, risks, assumptions, open questions and glossary updated to the vocabulary. OQ-6 resolved (Land is a segment). Success metrics and stack unchanged. See Section 17.

1. Problem Statement

Real estate market data reaches brokers through many unrelated channels: newspaper classifieds, WhatsApp groups, property websites, enquiry forms, Instagram and Facebook posts, developer price sheets in Excel, and databases shared between brokers. Each source uses its own layout, column names, spelling and level of completeness.

Today this data is:

- **Scattered** across files and chats, so nobody has a single view of what is available (supply) and what is wanted (demand).
- **Duplicated**. The same ad runs for days and in more than one newspaper. The same property appears in more than one broker sheet. The same client reaches the firm through a broker, an ad and a referral. This inflates volume, wastes follow up effort and creates commission disputes.
- **Unstructured**. Free text such as "2BHK Andheri W 1.2Cr negotiable, call 98xxxxxx" must be read by a person to be understood.
- **Unverified and ageing**. Most records are never checked with the source, and nobody tracks when a property was last confirmed as available. Listings stay up long after the property is sold or rented, and old requirements stay open after the client has moved on.
- **Manually matched**. Pairing a requirement with a property depends on a broker's memory, so good matches are missed. When nothing matches, the search for new supply on portals and through the network happens outside any system.

- **Without a working process.** There is no shared rule for when to call an owner, when to publish, when to follow up a client or when to let a requirement go. Deals stall between site visit and closing because nobody owns the follow up.
- **Hard to query or publish.** Answering "how many 3BHK rentals under ₹1L in Bandra?" or publishing inventory to the company website needs manual spreadsheet work.

2. Business Context and Users

Phase 1 is an internal tool for 11 Estates, run by two teams: a **demand team** that works buyers and tenants, and a **supply team** that works owners, brokers and developers. Later, the same product is sold as SaaS to other brokers, where each customer (tenant) sees only its own data.

User	Who they are	What they need from the system
Demand team	11 Estates staff who work buyers and tenants from first contact to closing	A queue of demand to contact, qualify, reconfirm and follow up; matched supply; proposals, site visits and In process deals in one place; a way to request supply when nothing matches
Supply team	11 Estates staff who work owners, brokers and developers	A call queue ranked by priority and daily capacity; tools to verify, publish and reconfirm offers; incoming sourcing requests from the demand team
Owner / manager	Runs the business	Demand and supply dashboards: queues, life curve health, exits, conversion and source quality
Data operator	Staff who collect and upload files (may sit in either team)	Upload any sheet in any template quickly; see what was accepted, rejected and duplicated
Listing consumer (system)	The 11 Estates website and project microsites	A stable API returning what may be published, at the right publication level
(Future) Tenant admin	Admin at a broker firm that buys the SaaS	Manage their own users and data, isolated from other tenants

3. Business Goals

- **G1: One clean database.** Every record lands in one place in a common structure (Properties, Projects, Offers, Demand, People and Enquiries), with duplicates and reposts identified across supply, demand and people.
- **G2: Automatic understanding.** Every record is classified by record_scope, deal_type, market, segment, property_type and side, with source type, using one controlled vocabulary shared with the extractors, without manual tagging.
- **G3: Automatic matching (core value).** The system continuously pairs demand with supply, many to many, respecting location, area, price and availability dates, and ranks the matches.
- **G4: Guided journeys.** Every offer and every demand moves through defined stages. Queues tell each team what to do next, and a life curve retires records nobody has confirmed.
- **G5: Self serve insight.** Demand and supply dashboards, plus natural language chat over the data that answers questions, shows tables and produces Excel downloads.
- **G6: Distribution.** Publishable supply, project microsites and anonymous demand posts are served through a listings API, within the publication rules.
- **G7: SaaS ready.** Built for one company now, in a way that lets the product be sold to other brokers later with strict data separation and no re-architecture.

4. Data Model and Record Rules (business view)

4.1 Core records

Record	ID	What it is
Property	PRP	A physical asset: flat, office, shop, warehouse, plot or whole building. Carries segment, property_type, property_detail, land_use (Land only), areas, location (locality, city, state, landmark) and photos.
Project	PRJ	A new development sold by its developer (market = Primary). Holds developer, RERA registration and unit configurations, each described by property_type with bhk_min and bhk_max.
Offer	INV	One deal on a Property, or one unit configuration in a Project with a unit count. Each Offer has exactly one deal_type and its own price fields (sale_price_inr_min, sale_price_inr_max, rent_monthly_inr_min, rent_monthly_inr_max, current_rent_inr, deposit_inr), deal tags, status, life curve and matches. A supply record offered as Sale Lease is one Property with two Offers.
Demand	DEM	A requirement with side = Demand. Carries deal_type (one or more), market (Primary, Secondary or Any, on Sale only), segment, property_type (one or more), budget and size as _min and _max fields, and any deal tags the client states.
Person		A person or phone number. party_type (Owner, Broker, Developer, Company, Bank, Society, Government) says who they are, independent of side. One person links to many records and can be on both sides at once, for example selling one flat to buy another, with the dependency recorded. Market Participants are People with a participant_role.
Enquiry	ENQ	A response from the market found in an uploaded export, tied to the offer, campaign or project that produced it.
Touch		One arrival of a demand through a source. The first touch gets the credit.
Match	MAT	A ranked pairing of a Demand with one Offer or a bundle of Offers.
Sourcing request	SRQ	The demand team's request to the supply team to find supply for a demand with no match.
Proposal / Site visit / Deal	PROP / DEAL	Options shared with a client; a client visit to an offer; one demand and one offer from agreed terms through In process to Closed or cancelled.
Business	BIZ	An operating business offered or wanted. Carries deal_type, side, sector, includes_property and business_description. When includes_property = Yes, a linked Property record is created so the asset is visible to the Supply team.
Capital	CAP	Money sought or offered for a company or project, or a loan book sold. Carries deal_type, side, sector and business_description.
Equipment	EQP	Machinery or plant without premises. Carries deal_type, side and business_description.
Watchlist item	WCH	A Market Signal: signal_type, party_type and deadline_date, with a follow up task for the Supply team.
Raw record / sighting		A row exactly as uploaded; one raw appearance of a record.
Market data point		A closed or reported price kept as intelligence.

Every record also carries record_scope, side, side_evidence (extracted records), needs_review, review_reason, source type, capture mode and tenant ID. IDs for the Business, Capital, Equipment and Watchlist records are proposals for Stage 2.

4.2 Record classification and deal vocabulary

This standard is shared by the CRM, the newspaper extractor and the WhatsApp extractor. Upload files use these exact field names and values.

Classification principles

- **One fact per field.** Side, deal type, market, segment and property type are separate fields. "Resale, for sale", "Looking For Lease" or "Commercial Office" are labels or legacy values, never stored values.

- **Store canonical values, show generated labels.** deal_type = Lease is stored for both residential rent and commercial lease; deal_type = Sale for both sellers and buyers. The UI builds labels from stored fields (see display labels below).
- **Side is inferred at extraction.** Channel and Digi records get side from the ad text, with the deciding phrase kept in side_evidence. Direct records get side chosen by the user on the form.
- **Controlled values only.** Classification fields accept only the values in this section. Adding a value changes the standard and is owned by Vinit and Priyanka.
- **Blank means unknown.** No "Unknown", "NA", "Other" or "Various" placeholders in controlled fields. The one explicit non value is side = None for scopes with no buyer or seller.
- **Keep every record.** Nothing captured is discarded; records that are not property deals are kept and classified by record_scope.
- **Doubt is flagged.** Records the extractor cannot classify with confidence carry needs_review = TRUE and a review_reason, and appear in the review queue.

Classification order. Used by the extractors and by the manual entry form: (1) record_scope; (2) deal_type, from the list allowed for that scope; (3) market, Sale only; (4) segment, Property only; (5) property_type, from the list for that segment; (6) side, inferred last from the text and answers 1 to 5. The manual form may ask side first for usability, since the user knows whether they are entering inventory or a requirement; the stored fields are the same.

record_scope

Value	Meaning	Allowed deal_type	side	Routed to
Property	Land, a building or a space offered or wanted, even if furnished, equipped, tenanted or auctioned	Sale, Lease, JV, Pagdi	Supply, Demand	Supply Team or Demand Team
Business	An operating business offered or wanted	Sale, Lease, Partnership, Distribution	Supply, Demand	Business Desk
Capital	Money sought or offered for a company or project, or a loan book sold	Equity, Debt, Project Funding, Asset Sale	Supply, Demand	Capital Desk
Equipment	Machinery or plant without the premises	Sale, Lease	Supply, Demand	Archive
Market Participant	Someone advertising their own service (broker, auctioneer, architect, PMC, consultant, lender)	none	None	Network (People directory)
Market Signal	A notice pointing to a future deal (society hiring a PMC, government tender, land policy, title notice)	none	None	Watchlist, with a follow up task for the Supply team

Only Property records enter the Supply and Demand journeys (Sections 5 and 6). Business, Capital and Equipment records live in a list view per desk, Market Participants in the People directory, and Market Signals in the Watchlist. In Phase 1 these five scopes are stored and viewable, with no journey of their own.

deal_type for Property

Value	Meaning	Absorbs these older terms
Sale	Ownership transfer. Used on both the seller's and the buyer's record	Sale, Sell, Resale, Buy, Purchase, Acquisition, Outright
Lease	Any rental	Rent, Lease, Lease Out, Rent Out, Leave and License, lease transfer of premises
JV	Joint development or redevelopment between a landowner or society and a developer	JV, JD, Joint Venture, Joint Development, Redevelopment, Development Rights
Pagdi	Transfer of tenancy rights under Mumbai rent control. The incoming tenant pays a premium, pays nominal rent and never owns the property	Pagdi, Pagadi, Pagri, tenancy transfer

A supply record offered under two deals on the same asset stores both (Sale|Lease); in the data model this is one Property with two Offers. A demand record open to two deals also stores both.

market (Sale only)

Value	Meaning	Shown as
Primary	Sold by the developer	New Project
Secondary	Sold by an existing owner	Resale
Any	Demand only; the buyer has no preference	(omitted from the label)

New project is deal_type = Sale with market = Primary. A developer leasing out its own building is deal_type = Lease with market blank.

segment and property_type

segment	property_type values
Residential	Apartment, Penthouse, Studio, Villa, Bungalow, Row House, Farmhouse, Building, Serviced Apartment
Commercial	Office, Shop, Showroom, Restaurant Space, Commercial Building, Coworking, Hotel, Resort, Institutional Building, Commercial Space
Industrial	Gala, Shed, Warehouse, Factory, Industrial Building, Cold Storage
Land	Plot, Land Parcel, Agricultural Land

- Land is the fourth segment (OQ-6 resolved). Land records also carry land_use: Residential, Commercial, Industrial, Agricultural, NA, Mixed.
- Hotels and resorts sold or leased as premises are Commercial property types. Hotels sold as running businesses are Business scope.
- Building under Residential means a whole building or society, used for redevelopment and "entire building" requirements.
- Descriptive words (Luxury, Duplex, Jodi) go in property_detail or tags, never in property_type. Several types in one record are pipe separated (Office|Showroom).

side

Value	Meaning
Supply	The party has something to offer
Demand	The party is looking for something
None	Market Participant and Market Signal only

side_evidence stores the phrase that decided side for extracted records. A blank side on any other scope requires needs_review = TRUE.

Deal tags (attributes, never deal types)

Field	Values
sale_mode	Private, Auction
deadline_date	Date: auction, tender, offer or EOI last date
tenancy_status	Vacant, Tenanted (Tenanted on a Sale is a preleased investment; the tenant's rent is current_rent_inr)
tenure	Freehold, Leasehold (MIDC and CIDCO plots are leasehold)
agreement_form	Leave and License, Registered Lease
is_jodi	TRUE, FALSE
possession_status	Ready, Under Construction, Under Redevelopment, Available From
possession_date	Date or month; this is the Available from date used by the journeys and matching

Field	Values
furnishing	Furnished, Semi Furnished, Unfurnished, Bare Shell
price_negotiable	TRUE, FALSE

Party fields. party_type is independent of side. Party roles used in labels and conversation: Seller and Buyer (Sale); Developer and Buyer (Sale, Primary); Landlord and Tenant (Lease); Landowner or Society and Developer (JV); Outgoing and Incoming tenant (Pagdi). "Lessor" and "lessee" are not used.

Non property fields

Field	Applies to	Values
sector	Business, Capital	Hospitality, Education, Manufacturing, Food and Beverage, Healthcare, Media, Distribution, Agriculture, Technology, Real Estate, Other
includes_property	Business	Yes, No, blank
participant_role	Market Participant	Broker, Developer, Auctioneer, Architect, PMC, Consultant, Lender
signal_type	Market Signal	Redevelopment Upcoming, Government Tender, Land Policy, Title Notice

Display labels. Labels are generated, never stored. Supply labels start with "For"; Demand labels start with "Wants". Rent is how Lease reads for Residential. Filters and reports work on stored fields (deal_type = Lease), never on label words.

deal_type	Supply label	Demand label	Parties
Sale, Secondary	Resale, For Sale	Wants to Buy, Resale	Seller and Buyer
Sale, Primary	New Project, For Sale	Wants to Buy, New Project	Developer and Buyer
Sale, Any (demand)		Wants to Buy	Buyer
Lease, Residential	For Rent	Wants to Rent	Landlord and Tenant
Lease, Commercial, Industrial, Land	For Lease	Wants to Lease	Landlord and Tenant
JV	For JV	Wants JV	Landowner or Society and Developer
Pagdi	Pagdi, For Transfer	Wants Pagdi	Outgoing and Incoming tenant

Supply and Demand records share the same deal_type, market, segment and property_type fields, so matching is symmetric.

Field naming convention. snake_case; units in the name (_inr, _sqft, _pct); _min and _max for ranges (budgets, sizes, BHK); _date for dates; booleans TRUE or FALSE. Sale price, asking rent and a tenant's current rent are separate fields. The full field dictionary is set in Stage 2 and matches the extractor upload columns one to one; Section 16 defines the terms.

4.3 Source types

Every record carries a source type. In Phase 1 records enter by file upload or manual entry; connectors come in the next phase.

Source type	Where it comes from	How it enters in Phase 1	How side is set
Channel	Newspaper classifieds, WhatsApp groups, property portals	Uploaded extractor files in the standard schema; portal finds typed in during sourcing are Channel, typed in	Inferred at extraction, with side_evidence
Digi	Instagram and Meta posts and ads, microsites, website enquiry forms, enquiries on 11 Estates listings	Uploaded exports of lead forms and enquiry logs, each row carrying the campaign, form, project or listing ID	Inferred at extraction, with side_evidence
Direct	References, direct calls, messages and walk ins to the team	Typed in through quick add	Chosen by the user on the form

Source type and capture mode are stored separately (for example "Channel, typed in"). When one demand arrives through more than one source, each arrival is stored as a touch and reporting credits the first touch.

4.4 Status model

Each Property record has independent status axes, so an offer can be unverified and still published anonymously, or matched while its record is only Captured.

Supply offer axis	Stages	Meaning
Record	Captured, Enriched, Contacted, Verified, Qualified	What the team knows. Enriched covers automatic and desk work. Contacted means a person spoke to the source. Verified means real photos and details are confirmed. Qualified means terms are agreed with the source.
Publication	Private, Anonymous, Public	What the market can see, capped by the rules in 4.6.
Commercial	Upcoming, Available, Matched, In proposal, Site visit, In process, Closed, Inactive	Deal progress. Upcoming means possession_status = Available From with a future possession_date. Inactive means gone, sold or leased elsewhere, or the source is unwilling.
Signals	Enquiries, matches, sightings	Market interest. Signals raise call priority. They are counters on the record, separate from its stages.

Demand axis	Stages	Meaning
Record	Captured, Enriched, Verified, Qualified	What the team knows about the requirement and the client.
Publication	Private, Anonymous	Anonymous demand posts are used during sourcing and show "Wants" labels.
Commercial	New, Contacted, Active, Sourcing, Matched, Proposal shared, Site visit, In process, Closed	Sourcing is skipped when the CRM already holds a match. In process covers negotiation, contract work and registration.
Exits	Lost, Dormant, Invalid	Lost records the reason and any competing terms. Dormant records a revisit date and pauses the life curve. Invalid records the reason and flags the person.

Matches and deals can move backwards. When a token is refunded, a loan is rejected or a landlord withdraws, the offer returns to Available, the demand returns to Active, and the failed deal stays on record.

4.5 Life curve

Every live offer and every live demand counts the days since it was last confirmed on a call, meeting or visit. Any confirmation resets the count to day 0. Reposts and new sightings update **last seen** and leave **last confirmed** unchanged. For Sale, Primary offers the count follows the date of the developer's latest price sheet. For offers with a future possession_date, the count starts 60 days before that date.

Supply offer	Fresh	Ageing	Stale	Expired
Lease, Residential	to 14 days	to 30	to 45	after 45
Lease, Commercial	to 30 days	to 60	to 90	after 90
Sale, Secondary	to 45 days	to 90	to 120	after 120
Sale, Primary (projects)	to 30 days	to 60	to 90	after 90
Industrial, Land, JV, Pagdi	OQ-19			

Demand	Fresh	Ageing	Stale	Expired
Lease, Residential	to 14 days	to 21	to 30	after 30
Lease, Commercial	to 30 days	to 60	to 90	after 90
Sale, Secondary or Any	to 45 days	to 90	to 150	after 150
Sale, Primary	to 30 days	to 60	to 120	after 120

Demand	Fresh	Ageing	Stale	Expired
Industrial, Land, JV, Pagdi	OQ-19			

Stage	Rule for a supply offer	Rule for a demand
Fresh	All publication levels allowed	No action
Ageing	Added to Should call for reconfirmation	Reconfirm call added to the demand team's queue
Stale	Public drops to Anonymous automatically; moves up Should call	New match suggestions stop until the client reconfirms
Expired	Unpublished everywhere, including anonymous listings; left out of automatic matching until reconfirmed	Moves to Dormant automatically

Thresholds are initial values for confirmation (OQ-19).

4.6 Publication rules

- **Private** is always allowed.
- **Anonymous** is allowed while an offer is Fresh or Ageing, including before anyone has spoken to the source. It shows locality, area, price and availability, with a note that details are subject to confirmation.
- **Public** is allowed only when the offer is Verified with real photos and is Fresh or Ageing. Photos shared by the source count as permission; there is no separate consent step. If in doubt, the team keeps the offer Anonymous.
- The rules set a ceiling. The team decides where to publish within it.
- Listings show generated labels (4.2), never stored values or legacy terms.
- No publication level exposes contact details or the exact address (building or society name, wing, flat number or street), including details hidden in free text or photos. Building names go only into proposals for qualified clients.
- Every listing shows the 11 Estates MahaRERA agent registration number. Sale, Primary listings and microsites also show the project's RERA registration number.
- Closed, Inactive and Expired offers are unpublished automatically. Upcoming offers may be published with their possession_date.

4.7 Intake validation and review queue

- Newspaper and WhatsApp extractor files arrive in the standard schema, with the exact column names and values of 4.2.
- Uploads containing values outside the controlled lists are rejected row by row, with a clear error naming the field and value.
- Rows with needs_review = TRUE are loaded and appear in a review queue, grouped by review_reason. Review does not block routing.
- Manual entry forms follow the classification order. Dropdowns show only controlled values. market appears only when Sale is selected. property_type options filter by the chosen segment. Deal tags are optional fields.

5. Supply Journey

The supply team owns every offer from capture until it is matched. From the match onward, the demand team runs the client side while the supply team keeps the source informed and sees progress on the offer.

Stage	What happens	Owner
Capture	A row is uploaded and kept raw. Attributes are extracted. The duplicate check works at property level (building, floor, area, locality), never on phone number alone. Rows arrive with record_scope, side and deal fields; non Property records go to their desk, directory or Watchlist, and needs_review rows also appear in the review queue. It creates a Property and one Offer per deal_type, or links the row to an existing one as a sighting or a second source with any price gap flagged.	System
Queue	The offer enters Should call with a rank.	System
Publish anonymously	Optional, when the team has not yet spoken to the source.	Supply team
Signal	An enquiry or a match moves the offer to Must call.	System
Contact	The call confirms availability, price and areas and resets the life curve. Further offers on the same property (for example a Sale offer alongside a Lease offer) are added here.	Supply team
Verify	Real photos and details collected. Public unlocks.	Supply team
Qualify	Terms agreed with the source, including revenue share with a partner broker. The broker's own record is qualified.	Supply team
Match and hand off	Matched demands go to the demand team for proposal, site visit and In process.	Both
Close or retire	Closed: unpublished, other matched demands notified, closing price kept as market data. Inactive: kept for intelligence.	Both

5.1 Situations the journey must handle

Situation	How it runs
Interest first	No spare bandwidth. The offer waits anonymously until an enquiry or match moves it to Must call, then is contacted, verified and published Public.
Proactive vetting	Bandwidth is available, so the team calls before any interest, verifies and publishes Public directly.
Sourced for a demand	Created from a demand or sourcing request, starting at Contacted, tagged Sourced for DEM, matched on creation and verified at top priority. May stay Private while the client decides.
Ageing record	Nobody reconfirms. The life curve moves it through Ageing (reconfirm), Stale (downgraded to Anonymous) and Expired (unpublished). A reconfirm call resets it or retires it to Inactive.
Future availability	Created as Upcoming with possession_status = Available From and a possession_date; published with that date; matched only to demand whose timing fits; the life curve starts 60 days before the date. Closing an 11 month lease creates an Upcoming offer for the same property at month 10.
Sale, Primary (project)	A Project with unit configuration offers and unit counts, built from the developer's price sheet. Price revisions update offers and flag matches that go over budget. A booking reduces the unit count while the offer stays live.
Duplicate post	Another broker posts the same property. It is linked as a second source on the existing offer, with the price gap flagged.

5.2 Call queues and bandwidth

- **Must call** holds offers with an enquiry or a match. Target: contact within 24 hours of the enquiry being uploaded.
- **Should call** holds proactive vetting and reconfirmation, ranked by freshness, demand gaps (categories where open demand exceeds matching supply), source quality and price band.
- The team sets a daily call capacity. The day's list fills with Must call first and tops up from Should call.
- Every call records an outcome: confirmed; no answer (retry up to 3 attempts); already gone (Inactive, price kept as market data); unwilling to work with 11 Estates (kept for matching intelligence, never published).

6. Demand Journey

Stage	What happens	Owner
Capture	Channel and Digi demand is created from uploaded rows with side inferred and side_evidence kept. Direct demand is typed in through quick add, which looks up the phone number first, then asks side, deal_type, market, segment and property_type.	System / demand team
Deduplicate	The same phone number, company or requirement arriving through another source is merged as a touch on the existing demand. The first touch keeps the credit. Uncertain cases go to review.	System
Understand	First call: requirement, budget, timing, locations, decision maker. Resets the life curve.	Demand team
Qualify	Decision maker reached, budget and timing confirmed, agreement to work with 11 Estates.	Demand team
Check inventory	Decision point: does anything in inventory match? Yes leads to Match. No leads to Sourcing.	System
Match	The team confirms suggested matches, including bundles.	Demand team
Sourcing	A sourcing request goes to the supply team and the demand is posted anonymously. Supply found on portals or through the network is added through Add supply on the demand, prefilled from it and matched on creation, then verified at top priority. The post comes down once matched.	Both teams
Proposal shared	Options with photos, built up and carpet area, price, availability and building names.	Demand team
Site visit	Client visits; the visit resets both life curves.	Demand team, supply team
In process	Negotiation, contract work, stamp duty and registration. Every In process deal carries a next action and a follow up date; overdue deals rise to the top of the demand team's queue.	Demand team
Closed	Deal recorded. Other options go back into the pool. Other demands matched to the closed offer are notified and return to matching.	Demand team

6.1 Exits

Exit	When	What the system does
Lost	The client closes elsewhere or withdraws, at any stage	Records reason and competing terms as market data; releases all matches
Dormant	The client postpones, or the demand life curve expires	Records a revisit date; pauses the life curve; releases matches; returns the demand to the queue on the revisit date
Invalid	Fake or unreachable details, or a broker posing as a client	Records the reason; flags the person so future records from them are marked; drops out of all queues

6.2 Handoffs between the teams

- Sourcing request from demand team to supply team, with due date and priority.
- Match notification to the demand team when an offer matches an open demand.
- Proposal, visit and In process status visible on the matched offer for the supply team.
- Automatic notice to every other matched demand when an offer closes, and to matched offers when a demand exits.

7. Matching Rules (business level)

- Pairs opposite sides on the shared fields: deal_type (any of the demand's values), market (Any matches Primary and Secondary), segment and property_type (any of the demand's values); deal tags the client states (for example tenancy_status = Tenanted for a buyer of preleased property); micromarket, using a Mumbai locality hierarchy (for example Andheri East containing Chakala, Marol, MIDC); area, with built up and carpet stored separately and compared like with like; price or budget normalised to one unit; and the offer's possession_date against the demand's move in timing.
- **Many to many.** One offer can hold many demands and one demand many offers. Each pairing is its own match record.
- **Bundles.** Two or more offers can be matched together when only their combined area meets the requirement.
- **Date aware.** Offers whose possession_date falls after the demand's window are excluded with the reason recorded.
- Expired offers are excluded. Stale offers are shown with a reconfirm flag.
- When an offer closes, its other matches close with the reason "Leased to another client" or "Sold to another client", and those demands return to matching.
- When a price changes, matches where the new price exceeds the budget are flagged for review.
- Matches are suggestions. The team confirms them, and nobody is contacted automatically. Weights are set in Stage 2 using broker feedback (M6).

8. Dashboards (business view)

Each dashboard tile opens the underlying list. Dashboards group and filter on stored fields and show generated labels; Lease is one count, split by segment where useful. Chat (Section 9, Module B) answers questions beyond the tiles.

Demand dashboard	Contents
Demand by source	Channel, Digi and Direct counts; duplicates merged this week
Demand life curve	Fresh, Ageing, Stale and Expired counts
Demand team queues	To contact, To qualify, Reconfirm due, In sourcing, Sourcing requests open, Open matches, Proposals out, Site visits this week, In process with follow up due, Deals this month
Exits	Lost, Dormant and Invalid this month, with reasons
Demand by classification	Grid of segment (Residential, Commercial, Industrial, Land) by deal_type (Sale split by market Secondary, Primary, Any; Lease; JV; Pagdi), shown with Wants labels

Supply dashboard	Contents
Stock	Properties, Projects and Offers counts; duplicate posts linked this week
Supply life curve	Fresh, Ageing, Stale and Expired counts
Supply team queues and listings	Must call, Should call, Upcoming offers, share of offers verified, listed Public, listed Anonymous
Supply by classification	Offers by segment and deal_type (Sale split by market; Lease; JV; Pagdi), shown with For labels; deal tag filters such as sale_mode = Auction or tenancy_status = Tenanted

Other scopes	Contents
Business Desk and Capital Desk	Counts by deal_type, side and sector; Business records with includes_property = Yes and their linked Properties
Archive	Equipment records by side
Network	Market Participants by participant_role
Watchlist	Market Signals by signal_type; deadlines in the next 14 days; open follow up tasks

Data quality dashboard	Contents
Uploads	Files and rows per source, accepted, rejected with reasons (including values outside controlled lists), duplicates found, time since last upload per source
Review queue	needs_review records by review_reason; uncertain merges; flagged price gaps
Side checks	Records where side was defaulted to Supply, per run; a rise means the extractor phrase list needs new entries
Source quality	Share of records per source that verify, match and close

9. Solution Overview: Modules (Phase 1)

Module A: Ingestion and record entry

- Upload Excel/CSV files from any source, including exported WhatsApp chats, lead form exports and enquiry logs. A single file can have about 100,000 rows, with up to 10 uploads per day.
- Files arrive in many templates. The system maps each file to a common structure and remembers mappings for templates it has seen before.
- Raw data is always kept exactly as uploaded, with source, uploader and date, so every clean record traces back to its original rows.
- Processing runs in the background with a progress view and a per file report (rows accepted, rejected with reasons, duplicates found).
- Extractor files arrive in the standard schema; values outside the controlled lists are rejected row by row and needs_review rows go to the review queue (4.7).
- Manual entry and editing in the app, with forms following the classification order (4.7): quick add for Direct demand (phone lookup first), Add supply from a demand or sourcing request, and updates after calls and visits (outcome, availability, price, photos).
- Photo attachment to Properties and Offers.

Deduplication (Modules A and B)

- A phone number identifies a Person. It never identifies an Offer or a Demand on its own; one broker's number can advertise many properties.
- Reposts of the same ad are one record seen many times. Repeat sightings update "last seen".
- Supply dedup works at property level, and extra sources on the same property are linked as second sources.
- Demand dedup matches phone number, company and requirement across sources and merges arrivals as touches.
- Outcome rules: raw kept in full; each real property, offer, requirement and person appears once, linked to its sightings; uncertain cases go to human review and are never merged silently; every merge can be undone.
- Exact matching rules are worked out in Stage 2/3 from sample files (OQ-4).

Module B: Intelligence and journeys

- **Classification** in the order of 4.2 (record_scope, deal_type, market, segment, property_type, then side with side_evidence), with attributes pulled from text into the named fields (locality, building, property_type, bhk_min and bhk_max, areas, price or rent, deal tags). Low confidence records carry needs_review and a review_reason.
- **Matching engine** following Section 7.
- **Journeys and queues** following Sections 4 to 6: status axes, life curve engine, Must call and Should call, demand queues, sourcing requests, proposals, site visits, In process follow ups and exits.
- **Dashboards** following Section 8.

- **Chat interface.** Plain language questions answered with text, tables or an Excel file. Model and hosting decided in Stage 3.

Module C: Listings API

- Serves Property offers in all four segments to the 11 Estates website and project microsites, at their publication level (Anonymous or Public), plus anonymous demand posts.
- Exposed: generated label, photos (Public only), price or rent, deal_type, market, segment, property_type, bhk_min and bhk_max, area, locality, possession_date, deal tags such as sale_mode and tenancy_status, amenities, floor, furnishing, possession status, and the RERA numbers in 4.6. Exact field list set in Stage 2.
- Never exposed: exact address or contact details, including phone numbers or addresses hidden in free text or photos.
- Listings are withdrawn automatically when offers close, retire or expire, and downgraded when they turn Stale.

Delivery approach

User interface first (screens and flows) in Stage 2 (PRD). The backend roadmap and implementation plan follow in Stages 3 to 5.

10. Success Metrics (unchanged from 0.3; please confirm or change)

#	Metric	Proposed target	Measured how
M1	Upload success: share of uploaded rows parsed into the common structure without manual fixing	≥ 90%	Ingestion report per file
M2	Time from upload to categorised and searchable, for a 100,000 row file	≤ 30 min	System timestamps
M3	Dedup precision: share of auto merged records that really are the same listing	≥ 95%	Monthly manual sample of 200 merges
M4	Repost recall: share of repeated newspaper ads correctly linked to the existing listing	≥ 85%	Manual sample from a known newspaper batch
M5	Categorisation accuracy (Supply / Demand + segment)	≥ 90%	Manual sample review
M6	Match usefulness: share of top 5 matches a broker rates "relevant"	≥ 60%	In app feedback on matches
M7	Chat answer correctness on a fixed benchmark of business questions	≥ 85%	Benchmark set run at each release
M8	Privacy: listings served by the API that contain contact details or an exact address	0	Automated scan of API output plus audit sample
M9	Listings API live on 11 Estates website(s)	≥ 1 site	Integration count

11. Scope

In scope (Phase 1)

- 1 Upload of Excel (.xlsx/.xls) and CSV files in any template from any listed source, including WhatsApp chat exports, lead form exports and enquiry logs; about 100k rows per file, up to 10 files per day.
- 2 Template and column mapping with reusable saved mappings.
- 3 Preservation of raw uploads and row level lineage.
- 4 Manual entry and editing in the app: quick add, Add supply, and updates after calls and visits.
- 5 Deduplication of properties, offers, demand and people, including repost detection and demand touches, with a review queue and reversible merges.

- 6 Data model with Property, Project, Offer, Demand, Person (party_type, roles), Enquiry, Touch, Match, Sourcing request, Proposal, Site visit, Deal, Business, Capital, Equipment and Watchlist item.
- 7 Classification with the controlled deal vocabulary of 4.2 (record_scope, deal_type, market, segment, property_type, side, deal tags) and source type; extraction of attributes from free text.
- 8 Row level validation of uploads against the controlled lists, and a review queue for needs_review records.
- 9 Storage and list views for the five non Property scopes: Business Desk, Capital Desk, Archive, People directory (Network) and Watchlist, including linked Property records for Business with includes_property = Yes.
- 10 Status model, life curve engine and call queues for both teams.
- 11 Supply and demand journeys through Closed, with exits and In process follow up tracking.
- 12 Automatic many to many matching with ranking, bundles and date awareness.
- 13 Demand, supply and data quality dashboards.
- 14 Natural language chat over the data, with answers, tables and Excel download.
- 15 Listings API (read only) for the 11 Estates website and project microsites, with publication levels, photos and anonymous demand posts, excluding exact address and contacts.
- 16 Photo attachment for Properties and Offers.
- 17 City and locality reference data for Mumbai (MMR) with a micromarket hierarchy, designed so more Indian cities can be added.
- 18 Tenant aware data model from day one.
- 19 User interface design for all of the above (Stage 2).

Out of scope (Phase 1)

- Automated scraping or direct connectors to WhatsApp, Facebook, Instagram, newspapers, portals or website forms. Connectors are planned for the next phase; in Phase 1 records enter only by file upload or manual entry.
- Outbound communication from the system (SMS, WhatsApp, email). Proposals and demand posts shared in WhatsApp groups are sent by staff outside the system and logged.
- Commissions, invoicing, collections and accounting. Deals are tracked up to Closed.
- Document management and electronic signature.
- Building the website and microsites themselves (only the API that feeds them).
- SaaS commercial features: self sign up, tenant onboarding, billing, plans, tenant admin console. The data model is ready for them.
- Journeys for Business, Capital, Equipment, Market Participant and Market Signal records. They are stored and viewable only.
- Native mobile apps.
- Loading cities outside Mumbai/MMR.
- Steps for Sale, Primary (project) deals set aside for a later phase: registering clients with developers, developer empanelment, and booking to payout stages after Closed.

12. Constraints

Area	Constraint
Region	India only. Launch city Mumbai (MMR); more Indian cities later. INR (lakh/crore), sq ft (carpet and built up), BHK.
Compliance	Data includes personal names and phone numbers from third party sources, so the DPDP Act 2023 applies. Data stored in India (A-10). Listings show the 11 Estates MahaRERA agent registration number, and project listings and microsites show the project RERA registration number.

Area	Constraint
Freshness	With no connectors, enquiries reach the CRM only when exports are uploaded. The 24 hour Must call target counts from upload, so lead and enquiry exports need uploading at least daily (A-17).
Scale	About 100k rows per file, up to 10 files per day. Bulk and background processing required.
Tenancy	Single tenant in Phase 1. Architecture must allow strict multi tenant isolation later without re-architecture.
Architecture	Event driven microservices and the stage gated process in CLAUDE.md.
AI / model hosting	Open to Hugging Face and other open weight models. Decision in Stage 3. Inference cost at 100k row volumes is a real constraint.
Vocabulary	Classification fields accept only the controlled values in 4.2. Adding a value is a change to the standard, owned by Vinit and Priyanka, and applies to the CRM and both extractors together.
Budget / timeline / team	Not yet specified (OQ-11).

13. Key Risks

#	Risk	Why it matters	Early mitigation idea
R1	Personal data compliance. Names and phone numbers collected from WhatsApp groups, newspapers and shared broker databases without the individuals' consent.	DPDP penalties, reputation; worse once sold as SaaS	Confirm lawful basis; PII controls and access logging; never expose contacts via API; retention limits
R2	Template variety. Unpredictable layouts break parsing.	Low M1, manual work	Saved mappings, AI assisted column detection, review queue
R3	Free text quality. Abbreviations ("2bhk", "1.2 Cr", "85L"), Hindi/Marathi transliteration, inconsistent locality names ("Andheri W", "Andheri West", "Andheri (W)").	Poor categorisation, dedup and matching	Mumbai locality dictionary and micromarket hierarchy; unit normalisation; confidence scores; review queue
R4	Wrong dedup merges. Two different flats from the same broker marked as one, or two different clients merged as one demand.	Data loss, trust, wrong attribution	Phone number never the sole key; raw never destroyed; reversible merges; review thresholds
R5	Chat hallucination. Wrong answers stated confidently.	Bad decisions	Answers grounded in real query results; show the data behind each answer
R6	Stale supply and demand. Properties already sold or rented still listed; clients who moved on still chased.	Bad matches, bad listings, wasted calls	Life curve with category thresholds; automatic downgrade and unpublish; reconfirm queues; sightings as last seen
R7	AI inference cost at about 100k rows per file, up to 10 files per day.	Budget	Rules and normalisation first; AI only for rows rules cannot handle
R8	Privacy leak through listings. Phone numbers or building names inside free text or photos (for example a watermark).	Breaks the no contact, no exact address rule	Automatic sanitising before publishing; publication ceiling rules; team choice within the ceiling
R9	Future tenant data leak once sold as SaaS.	Critical trust and legal risk	Tenant ID on every record from day one; isolation enforced in every service
R10	Upload lag. Enquiries sit in exports until someone uploads them.	Slow response loses clients	Daily upload routine; time since last upload per source on the data quality dashboard; connectors in the next phase (OQ-7)
R11	Team bandwidth. Should call grows faster than the team can call.	Share of verified inventory falls; more stale listings	Daily call capacity; ranking by demand gaps; life curve retires records automatically
R12	Attribution disputes. The same client through a broker, an ad and a referral.	Commission disputes, broker trust	Demand dedup with touches; first touch rule (OQ-21); visible, reversible merges
R13	Follow up gaps after site visits.	Deals stall or are lost	In process stage with mandatory next action and follow up date; overdue queue

#	Risk	Why it matters	Early mitigation idea
R14	RERA non compliance on listings and microsites.	Penalties	RERA fields mandatory before publishing
R15	Wrong side. A requirement read as inventory, or the reverse.	Demand hidden as supply; wrong matches and listings	Side inferred last with side_evidence; needs_review when unclear; defaulted to Supply count tracked on the data quality dashboard

14. Assumptions

Any assumption you reject becomes a correction before approval.

#	Assumption
A-1	Currency is INR (lakh/crore notation). Units are sq ft (carpet and built up), BHK and acres.
A-2	Confirmed. Phase 1 has two ways in, both needed now: file upload (WhatsApp, newspaper, lead form and enquiry data arrives inside uploaded files) and manual entry and updates in the app (quick add, Add supply, updates after calls and visits). Connectors come in the next phase.
A-3	Phase 1 users are 11 Estates staff in a demand team and a supply team, plus management. No public sign up.
A-4	The Listings API is read only and consumed only by the 11 Estates website and project microsites.
A-5	Matching produces suggestions. It does not contact anyone automatically.
A-6	Raw data is kept unchanged. The clean view is separate and linked to raw rows. Whether dedup runs during upload or as a separate step is decided in Stage 3.
A-7	Data is mostly English, with some Hindi/Marathi transliteration.
A-8	Segments are Residential, Commercial, Industrial and Land. Property deal types are Sale, Lease, JV and Pagdi, with market (Primary, Secondary, Any) on Sale. The full vocabulary is in 4.2.
A-9	The user interface is a web application, desktop first.
A-10	All data is stored and processed in India (Indian cloud region).
A-11	Public listings show locality, area, price, availability and real photos. They never show the building or society name, wing, flat number or street address. Building names are shared only in proposals to qualified clients.
A-12	Up to 10 files of about 100k rows per day, so up to about 1M rows per day at peak.
A-13	Records that cannot be classified with confidence carry needs_review = TRUE and appear in the review queue. They are not deleted.
A-14	Photos shared by a source count as permission to publish them. There is no consent step.
A-15	Life curve thresholds in 4.5 are initial values.
A-16	The first touch of a demand receives the attribution credit.
A-17	Confirmed. Lead and enquiry exports are uploaded at least once a day.
A-18	Newspaper and WhatsApp extractor files use the standard schema: the field names and controlled values of 4.2.

15. Open Questions

None of these block approval of Stage 1. They will be settled in Stage 2 (PRD) unless noted.

#	Question	Needed by
OQ-4	Dedup rules. Please share 2 or 3 anonymised sample files, including one with a newspaper ad repeated over 3 or more days and one with the same client in two sources.	Stage 2
OQ-7	Connectors are planned for the next phase. Which source should get a direct connector first (WhatsApp Business API, website forms, Meta lead ads)?	Next phase planning

#	Question	Needed by
OQ-9	AI models: open weight (Hugging Face, self hosted or Inference Endpoints) or hosted APIs? Can personal data be sent to a third party AI provider?	Stage 3
OQ-10	User roles: do both teams see all data, or does each team see only its side plus matched records? Who works the Business Desk, Capital Desk and Watchlist?	Stage 2
OQ-11	Budget, target launch date, team size.	Stage 3
OQ-12	Matching weights, and whether the team is notified in app when a new match appears.	Stage 2
OQ-13	Please share about 10 real example questions. They become the M7 benchmark.	Stage 2
OQ-19	Confirm life curve thresholds, and set them for Industrial, Land, JV and Pagdi.	Stage 2
OQ-20	Daily call capacity for the supply team and the demand team.	Stage 2
OQ-21	Confirm the first touch rule, and how commission is shared when a broker introduced the client.	Stage 2
OQ-22	Proposal format: in app view, PDF or Excel export?	Stage 2
OQ-23	Number of call attempts before a source or client is marked unreachable (proposed 3).	Stage 2
OQ-24	Default revisit period for Dormant demand.	Stage 2
OQ-25	Confirm: hotels and resorts sit under Commercial; there is no Hospitality segment.	Stage 2
OQ-26	Confirm: 1 RK is stored as 0.5 BHK with property_type Studio.	Stage 2
OQ-27	Confirm: Pagdi is kept as a deal type.	Stage 2

Answered

#	Answer
OQ-6	Resolved in 0.6. Land is the fourth segment, with property_type Plot, Land Parcel or Agricultural Land, and a land_use field (4.2).
OQ-12 (part)	Matching compares opposite side, deal_type, market, segment, property_type, micromarket, normalised area, price or budget, and possession_date against move in timing (Section 7).
OQ-14	Records expire through the life curve, for both supply and demand (4.5).
OQ-15	Photos are collected on calls and visits or shared by the source, and attached to the Property and Offer.
OQ-16	Anonymous publishing is allowed while fresh; Public only once Verified with real photos; the team chooses within that ceiling (4.6).
OQ-17	Only the locality appears on the website. Building names go only into proposals to qualified clients (A-11).

16. Glossary

Term	Meaning
record_scope	What kind of opportunity a record is: Property, Business, Capital, Equipment, Market Participant or Market Signal
Side	Supply (has something to offer), Demand (looking for something) or None (Market Participant and Market Signal)
side_evidence	The phrase in the source text that decided side for an extracted record
Supply / Demand	A record with side = Supply / side = Demand. "Inventory" and "requirement" remain plain English words for them
deal_type	The kind of deal. Property: Sale, Lease, JV, Pagdi. Business: Sale, Lease, Partnership, Distribution. Capital: Equity, Debt, Project Funding, Asset Sale. Equipment: Sale, Lease
Sale	Ownership transfer, stored on both seller and buyer records
Lease	Any rental; shown as Rent for Residential

Term	Meaning
JV	Joint development or redevelopment between a landowner or society and a developer
Pagdi	Transfer of tenancy rights under Mumbai rent control; the incoming tenant pays a premium and nominal rent and never owns the property
market	On Sale: Primary (developer sells; shown as New Project), Secondary (existing owner sells; shown as Resale), Any (demand with no preference)
Primary / Secondary	See market
Segment	Residential, Commercial, Industrial or Land
property_type	The controlled type within a segment (4.2); descriptive words go in property_detail
land_use	For Land: Residential, Commercial, Industrial, Agricultural, NA, Mixed
Deal tags	Attributes of a deal, never deal types: sale_mode, deadline_date, tenancy_status, tenure, agreement_form, is_jodi, possession_status, possession_date, furnishing, price_negotiable
Jodi	Two adjoining units combined and sold together; is_jodi = TRUE
Preleased	A property sold with a tenant in place: deal_type Sale with tenancy_status = Tenanted
Leave and License	A rental agreement form common in Mumbai: deal_type Lease with agreement_form = Leave and License
party_type	Who a contact is, independent of side: Owner, Broker, Developer, Company, Bank, Society, Government
needs_review / review_reason	Flag and reason for a record the extractor could not classify with confidence
Property / Project / Offer	Physical asset / new development with unit configurations / one deal on a Property or one configuration in a Project, with its own deal_type and price
Listing	An Offer or demand post as published through the API, shown with a generated label
Enquiry	A market response found in an uploaded export
Source type	Channel, Digi or Direct
Touch	One arrival of a demand through a source
Raw record / Sighting / Repost	A row as uploaded / one raw appearance of a record / the same ad appearing again
Person / Contact	A person or phone number with party_type and roles
Match / Bundle	A ranked pairing of a Demand with an Offer / two or more offers matched together to one requirement
Life curve	Days since a record was last confirmed, with thresholds and automatic actions
Last confirmed / last seen	Last call, meeting or visit confirming a record / last sighting of it in an upload
Must call / Should call	Supply call queues: interest or match first; proactive vetting and reconfirmation when capacity allows
Sourcing request	The demand team's request to the supply team to find supply for a demand
In process	Deal stage after site visit covering negotiation, contract work and registration
Lost / Dormant / Invalid	Demand exits
Upcoming	An offer with possession_status = Available From and a future possession_date
Publication level	Private, Anonymous or Public
Business Desk / Capital Desk / Archive / Network / Watchlist	Where Business, Capital, Equipment, Market Participant and Market Signal records live
Tenant	A customer organisation. Phase 1 has one tenant (11 Estates)

Legacy terms (not used as stored values or field names)

Legacy term	Now
Transaction type	deal_type
Resale (as a type)	deal_type = Sale, market = Secondary
New project (as a type)	deal_type = Sale, market = Primary
Rent, Lease / Rent, Lease Out, Rent Out	deal_type = Lease; "Rent" survives only as the Residential label
Sale/Lease, Sale/Rent	deal_type = Sale Lease (one Property, two Offers)
Buy, Purchase, Acquisition	deal_type = Sale with side = Demand
JD, Joint Development, Redevelopment	JV
Listing category, asset class	segment plus property_type
Configuration (type and BHK together)	property_type plus bhk_min, bhk_max
Broker or owner	party_type
Builder	Developer
Pre-leased (as a type)	tenancy_status = Tenanted
Auction (as a type)	sale_mode = Auction
Lead	Deferred; Phase 1 needs are met by Enquiry
Unknown (as a value)	blank
Lessor, lessee	Landlord, Tenant

17. Change log entry: Deal vocabulary standardized

Version 0.6, 2026-09-24. Sections changed: 1, 3, 4 (4.1 rewritten; new 4.2 Record classification and deal vocabulary; 4.3 to 4.6 updated; new 4.7 Intake validation and review queue), 5, 6, 7, 8, 9, 11, 12, 13, 14, 15 (OQ-6 resolved; OQ-25 to OQ-27 added) and 16. Success metrics (Section 10) and stack unchanged.